



BINAY PowerLED Inspection Hand Lamps utilize a light source comprised of multiple high power white LEDs

- Very low power consumption of 10 watts or less
- Long life (over 50,000 burning hours), covered under 3-year warranty
- Unbreakable shock-proof and vibration-resistant construction; cool light output with no heat radiation
- Glare-free light output, focusing only on the work area



Binay Opto Electronics Pvt. Ltd.

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POWERING LED TECHNOLOGIES WORLDWIDE SINCE 1983

Get Light of 18W LED Tube in Just 12W Only!

Now Switch To 4-foot T-8

ULTRA HIGH EFFICIENCY PowerLED Tube Light

(Power Consumption 12W only; SAME light output)

33% Lower Power Consuming

MOST EFFICIENT POWER RATING LED TUBE LIGHTS IN THE MARKET!

While most comparative competitive LED tubes achieve a lumen output of 1600 lumens at 18W approx., the **BINAY ULTRA HIGH EFFICIENCY QUADRA-12-ES PowerLED Tube Light** is designed for 2000 source lumens at a 33% lower power consumption of 12W, thus achieving considerable power savings (apart from much longer maintenance free life). The light output performance is equivalent to that of the 1200mm T8 fluorescent tube.

5 YEARS Warranty WITH AN ESTIMATED LIFE OF 12-15 YEARS

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Wearable tech that can transform body heat into electricity

Researchers at North Carolina State University, USA, have developed prototypes known as wearable thermoelectric generators that harvest body heat into electricity, which can then be used to power devices as mobile phones.



NCSU researchers have developed wearables that harvest body heat and turn it into electricity (Image courtesy: NCSU)

The experimental new device is lightweight and conforms to the shape of the body, generating power by making use of temperature differential between the body and ambient air. The technology generates up to $20\mu\text{W}/\text{cm}^2$ of power without making use of a heat-sink, according to researchers.

The thermoelectric generator is a supply of renewable energy, which is limited only by your ability to move. Researchers incorporated the device into t-shirts where they found that the t-shirt thermoelectric generators were capable of generating $6\mu\text{W}/\text{cm}^2$, or as much as $16\mu\text{W}/\text{cm}^2$ of power when you are running.

Micro-sized, liquid-metal particles for heat-free soldering

Scientists at Iowa State University, USA, have developed micro-scale, liquid-metal particles that can be put to use for heat-free soldering, fabricating, repairing and processing of metals—all at room temperature. The project was initially undertaken as a way to stop liquid metal from returning to solid, even below the metal's melting point; something known as under-cooling in the field of metal structure and processing.

Martin Thou, assistant professor at Iowa State University, and his group have created liquid-metal particles containing field's metal (an alloy of bismuth, indium and tin) and particles containing an alloy of bismuth and tin. The particles are ten micrometres in diameter, about the size of a red blood cell. These were exposed to oxygen after which an oxidation layer was allowed to cover these, essentially creating a capsule containing the liquid metal. The layer was then polished until thin and smooth.